

Installation & Usage Guide

This document provides comprehensive, step-by-step instructions for the environment setup, execution, and testing of the **Mini Job Board** application across Windows, macOS, and Linux platforms.

Prerequisites

Before proceeding with the installation, ensure the following software components are installed and configured on your system:

- Java JDK 17 or higher: Recommended distributions include Eclipse Temurin.
- Node.js & npm: Version 16.x or higher is required; npm is typically bundled with Node.js.
- MySQL Server: Must be running locally on the default port 3306.

1. Database Setup

The backend infrastructure requires a dedicated MySQL schema titled `minijobboard`.

macOS (Using Homebrew)

1. Initialize the MySQL service: `brew services start mysql`.
2. Access the database terminal: `mysql -u root -p`.
3. Execute the creation command: `CREATE DATABASE minijobboard;`.

Windows

1. Install the official MySQL Installer from the developer website.
2. Launch the **MySQL 8.0 Command Line Client** from the Start Menu.
3. Authenticate using your root credentials.
4. Execute the creation command: `CREATE DATABASE minijobboard;`.

Linux

1. Authenticate to the MySQL instance: `mysql -u root -p`.
2. Execute the creation command: `CREATE DATABASE minijobboard;`.



2. Running the Backend API

To prevent conflicts with common development tools, the backend service is configured to run on port **8081**.

macOS & Linux

1. Navigate to the project directory: `cd backend`.
2. Grant execution permissions to the Maven wrapper: `chmod +x mvnw`.
3. Start the Spring Boot application: `./mvnw spring-boot:run`.

Windows (Command Prompt)

1. Navigate to the project directory: `cd backend`.
2. Start the Spring Boot application: `mvnw.cmd spring-boot:run`.



3. Running the React Frontend

The frontend development environment utilizes Vite and operates at **http://localhost:5173**.

1. Open a secondary terminal and navigate to the directory: `cd frontEnd`.
2. Install dependencies: `npm install`.
3. Launch the development server: `npm run dev`.



4. Verification & Testing

Backend Unit Tests

Validate business logic and constraints by executing the JUnit Mockito test suites:

- Command: `cd backend && ./mvnw test`.

Frontend Code Quality

Maintain code standards and prepare for deployment with the following scripts:

- **Linting:** `npm run lint` executes ESLint verification.
- **Production Build:** `npm run build` generates optimized assets in the `dist/` directory.